

1 状态变量、输入

状态变量45个

```
1 'Uc_DUT/CL'
2 'Uc_DUT/Cs'
3 'Uc_DUT/Cs1'
4 'Uc_DUT/Cs2'
5 'Uc_DUT/Cs3'
6 'Uc_DUT/Ct'
7 'Il_DUT/DUT/Lon'
8 'Uc_DUT/DUT/Series RLC Branch'
9 'Il_DUT/L1'
10 'Il_DUT/L2'
11 'Il_DUT/L3'
12 'Il_DUT/L4'
13 'Il_DUT/L5'
14 'Il_DUT/Va1/Lon'
15 'Uc_DUT/Va1/Series RLC Branch'
16 'Il_DUT/Va2/Lon'
17 'Uc_DUT/Va2/Series RLC Branch'
18 'Il_DUT/Va3/Lon'
19 'Uc_DUT/Va3/Series RLC Branch'
20 'Il_DUT/Va4/Lon'
21 'Uc_DUT/Va4/Series RLC Branch'
22 'Il_DUT/Va5/Lon'
23 'Uc_DUT/Va5/Series RLC Branch'
24 'Il_DUT/Vc1/Lon'
25 'Uc_DUT/Vc1/Series RLC Branch'
26 'Il_DUT/Vc2/Lon'
27 'Uc_DUT/Vc2/Series RLC Branch'
28 'Il_DUT/Vc3/Lon'
29 'Uc_DUT/Vc3/Series RLC Branch'
30 'Il_DUT/Vc4/Lon'
31 'Uc_DUT/Vc4/Series RLC Branch'
32 'Il_DUT/Vc5/Lon'
33 'Uc_DUT/Vc5/Series RLC Branch'
34 'Il_Series RLC Branch2'
35 'Il_Series RLC Branch3'
36 'Uc_T1/C'
```

```

37 'IL_T1/Lon'
38 'Uc_T2/C'
39 'IL_T2/Lon'
40 'Uc_T3/C'
41 'IL_T3/Lon'
42 'Uc_T4/C'
43 'IL_T4/Lon'
44 'Uc_T6/C'
45 'IL_T6/Lon'
'ILSeries RLC Branch1*'

```

输入变量23个（24:39均不使用）

```

1 'LDUT/Ideal Switch'
2 'LDUT/DUT/T'
3 'LDUT/Va1/T'
4 'LDUT/Va2/T'
5 'LDUT/Va3/T'
6 'LDUT/Va4/T'
7 'LDUT/Va5/T'
8 'LDUT/Vc1/T'
9 'LDUT/Vc2/T'
10 'LDUT/Vc3/T'
11 'LDUT/Vc4/T'
12 'LDUT/Vc5/T'
13 'LT1/T'
14 'LT2/T'
15 'LT3/T'
16 'LT4/T'
17 'LT6/T'
18 'U_DUT/DC Voltage Source'
19 'U_AC Voltage Source'
20 'U_AC Voltage Source1'
21 'U_AC Voltage Source2'
22 'U_DUT/AC Voltage Source1'
23 'IControlled Current Source'
24 'U_Vf: DUT/DUT/T'
25 'U_Vf: DUT/Va1/T'
26 'U_Vf: DUT/Va2/T'
27 'U_Vf: DUT/Va3/T'

```

28 'U_Vf: DUT/Va4/T'
29 'U_Vf: DUT/Va5/T'
30 'U_Vf: DUT/Vc1/T'
31 'U_Vf: DUT/Vc2/T'
32 'U_Vf: DUT/Vc3/T'
33 'U_Vf: DUT/Vc4/T'
34 'U_Vf: DUT/Vc5/T'
35 'U_Vf: T1/T'
36 'U_Vf: T2/T'
37 'U_Vf: T3/T'
38 'U_Vf: T4/T'
39 'U_Vf: T6/T'

2 转换矩阵

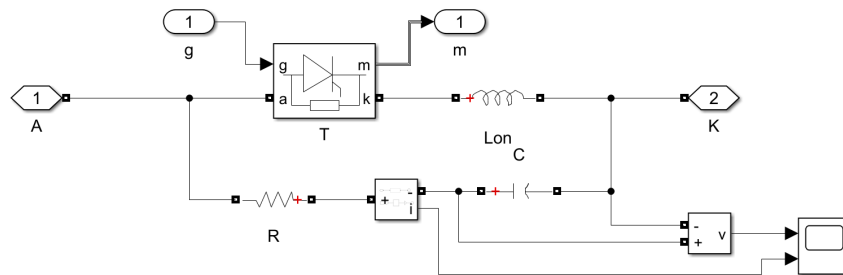


图 1: sim模型T1, T2, T3, T4, T6, Va1, Va2, Va3, Va4, Va5, Vc1, Vc2, Vc3, Vc4, Vc5

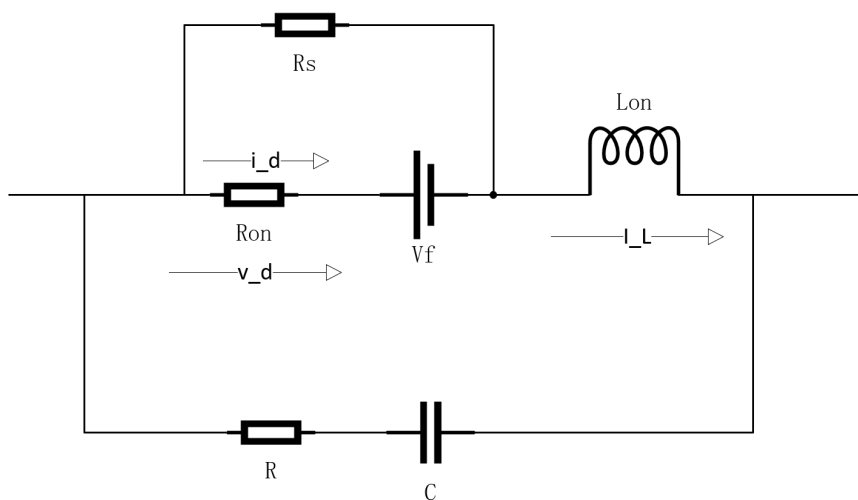


图 2: 等效电路T1, T2, T3, T4, T6, Va1, Va2, Va3, Va4, Va5, Vc1, Vc2, Vc3, Vc4, Vc5

state=1 时:

$$\left(\frac{1}{R_{on}} + \frac{1}{R_s}\right) \times v_d = i_L - \frac{V_f}{R_s}$$

$$i_d = \left(\frac{1}{R_{on}}\right) \times v_d$$

state=0 时:

$$\left(\frac{1}{R_s}\right) \times v_d = i_L - \frac{V_f}{R_s}$$

$$i_d = 0$$

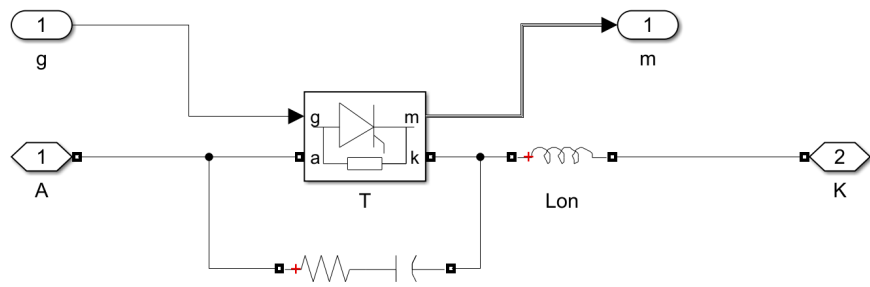


图 3: sim模型DUT

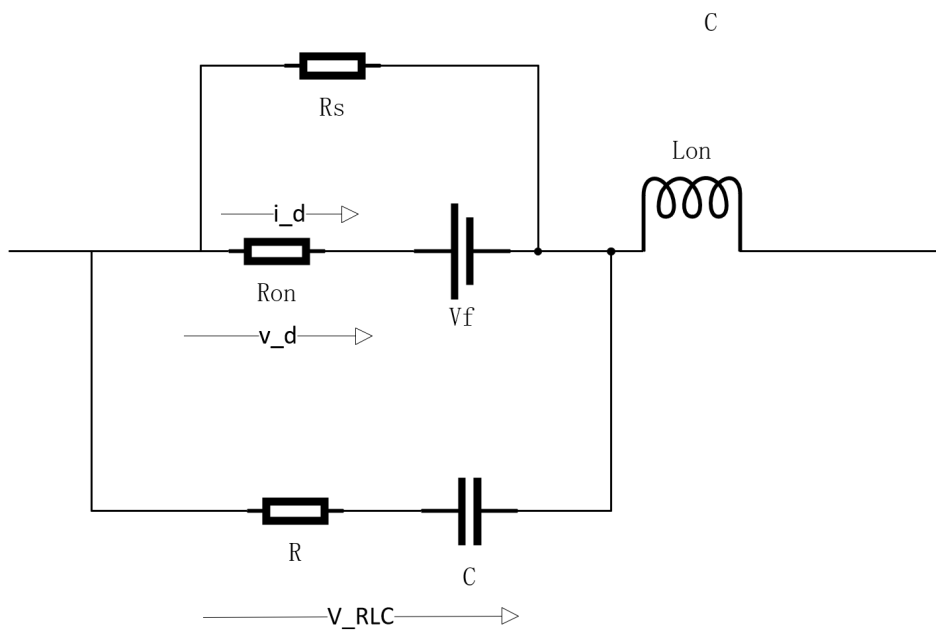


图 4: 等效电路DUT

state=1 时:

$$v_d = V_{RLC} - Vf$$

$$i_d = \left(\frac{1}{Ron}\right) \times v_d$$

state=0 时:

$$v_d = V_{RLC} - Vf$$

$$i_d = 0$$

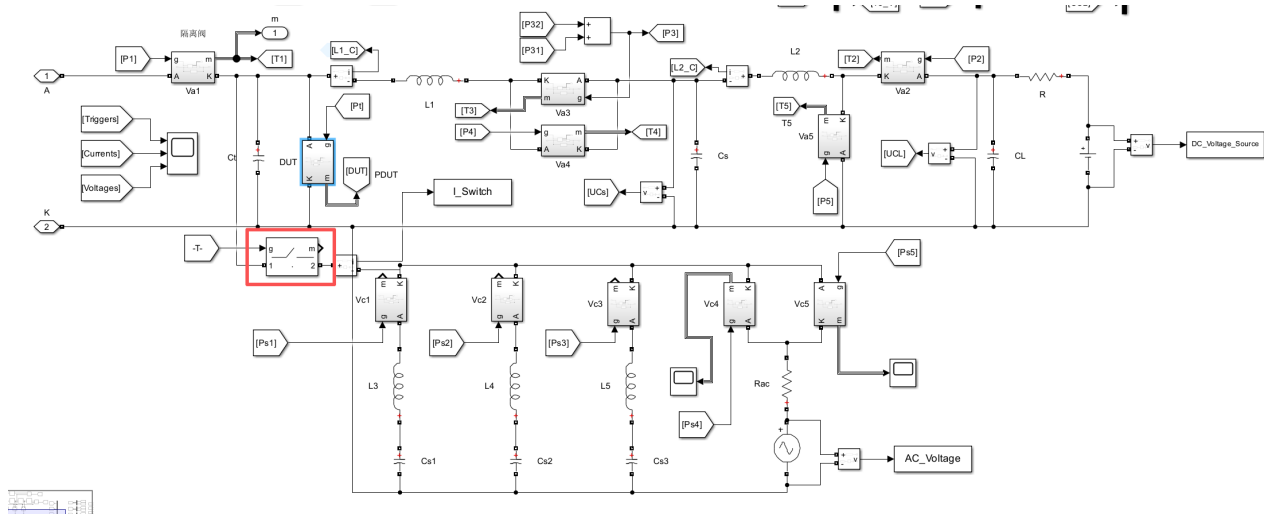


图 5: sim模型Ideal Switch

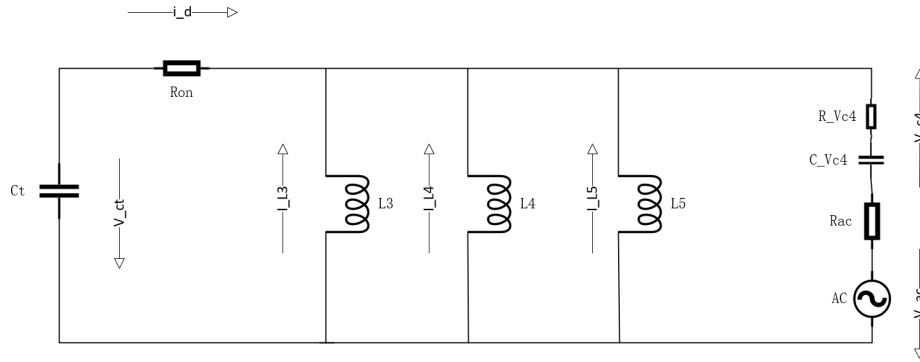


图 6: 等效电路Ideal Switch

state=1 时:

$$(1 + \frac{R_{ac}}{R_{on}}) \times v_d = V_{ct} + V_{c4} - R_{ac} \times i_{L3} - R_{ac} \times i_{L4} - R_{ac} \times i_{L5} - V_{ac}$$

$$i_d = (\frac{1}{R_{on}}) \times v_d$$

state=0 时:

$$v_d = V_{ct} + V_{c4} - R_{ac} \times i_{L3} - R_{ac} \times i_{L4} - R_{ac} \times i_{L5} - V_{ac}$$

$$i_d = 0$$

其中 V_{ac} 是已知源，所以从状态变量到输入的转换公式为 $A * v = B * x + b_{vf} + u_{vf} * u$

3 仿真结果

部分状态变量能够较好收敛。

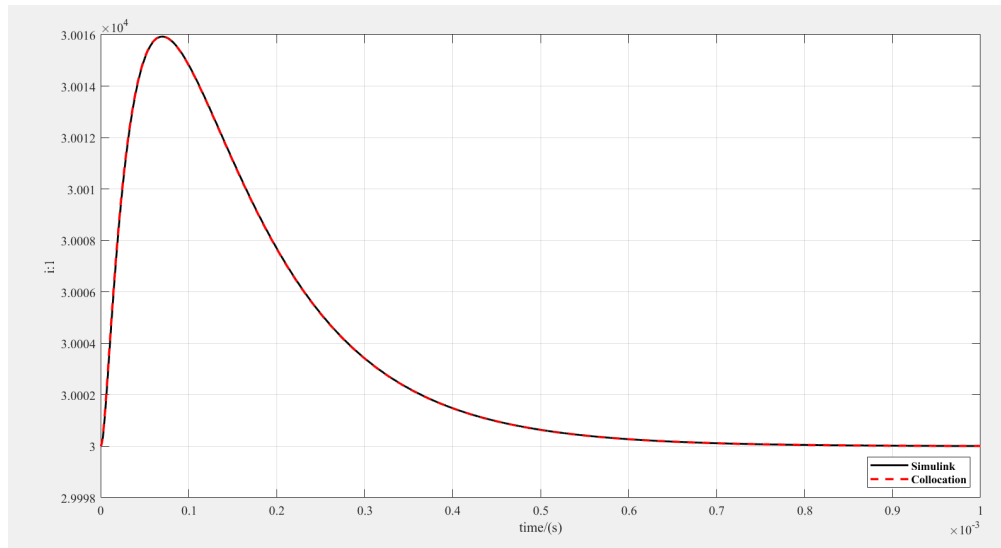


图 7: DUT-CL

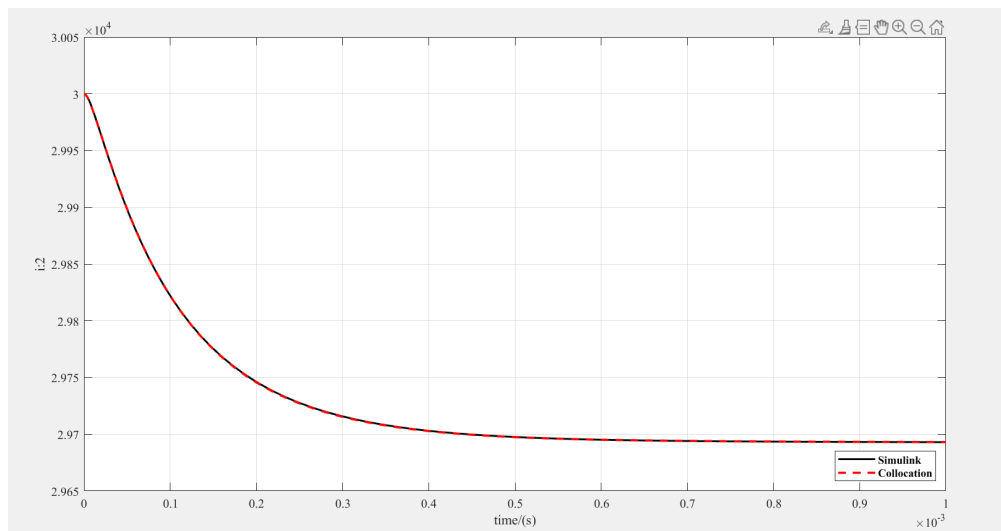


图 8: DUT-Cs

部分状态变量难以收敛

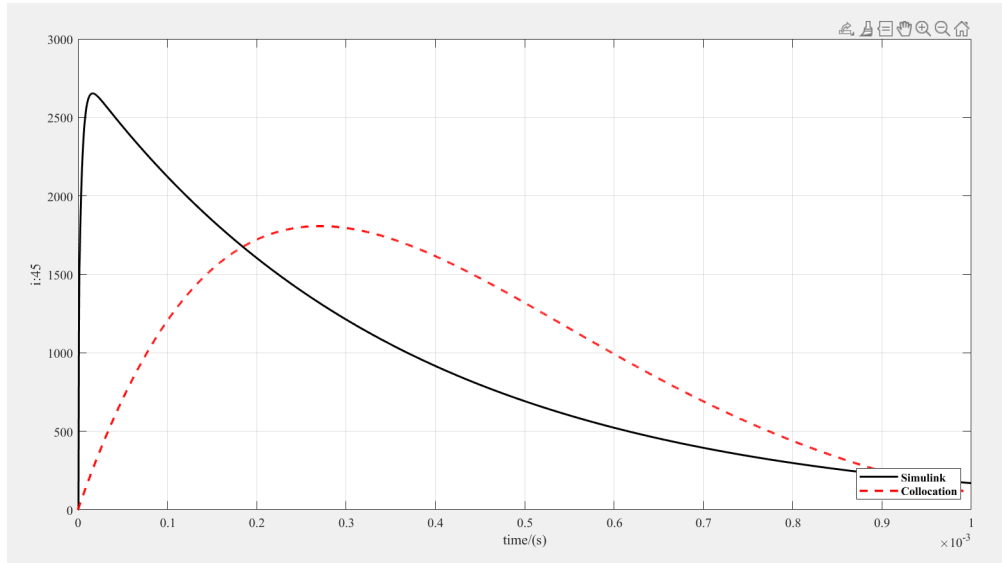


图 9: T6-IL ($k_{\max}=100$, $t=[0, 1e-3]$)

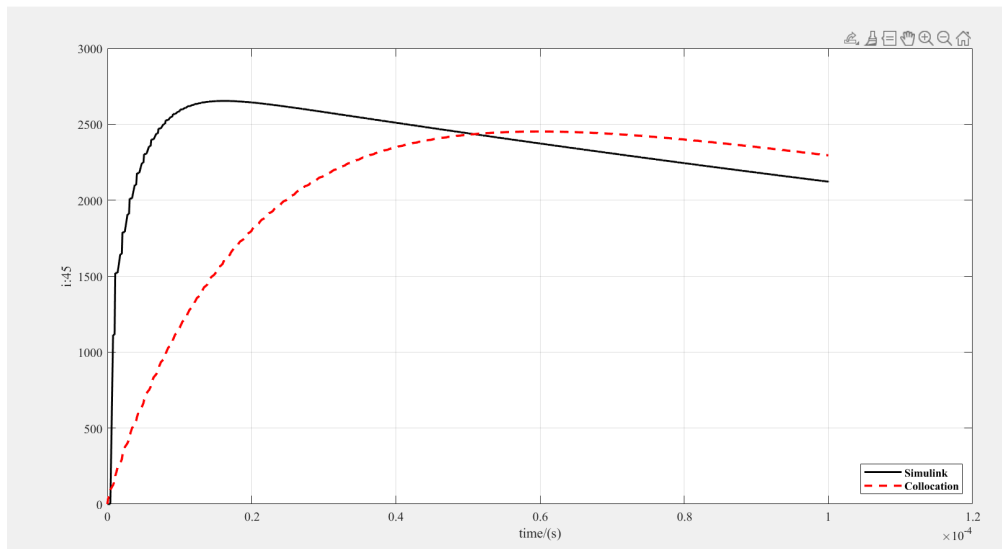


图 10: T6-IL ($k_{\max}=1000$, $t=[0, 1e-4]$)

Collocation运行时间($k_{\max}=1000$):

Collocation() Total Execution Time: 95.4388 s Collocation() Total Execution Steps: 101

Collocation_onestep(): 95.4328 s save_results(): 0.0012 s

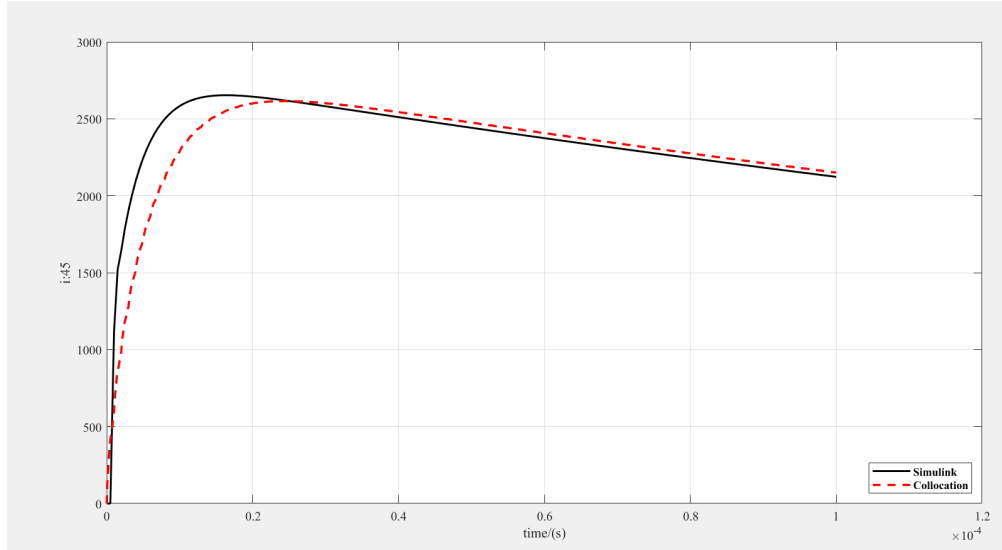


图 11: T6-IL ($k_{\max}=5000$, $t=[0,1e-4]$)

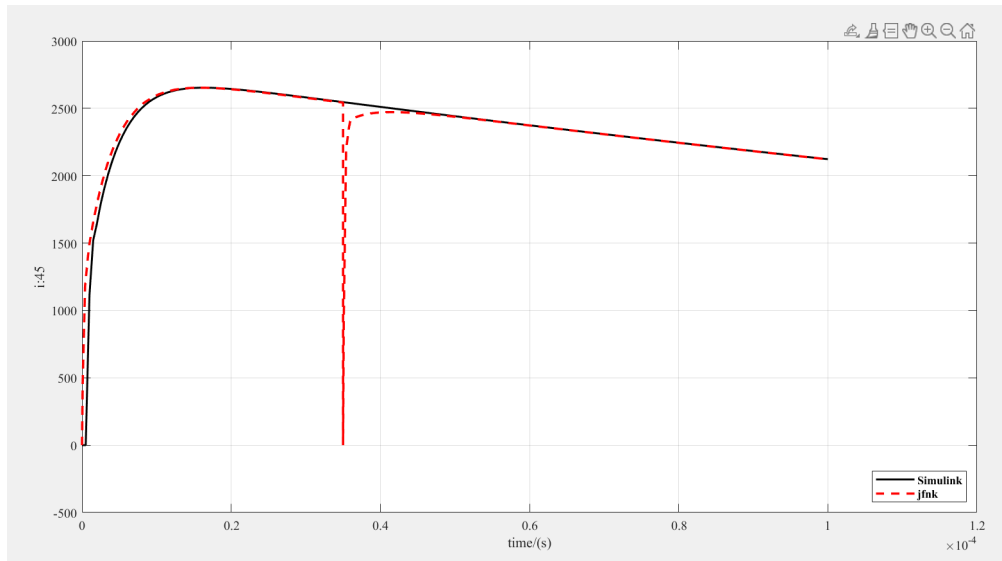


图 12: T6-IL (jfnk求解, $N_ITER_MAX_NEWTON = 50$, $N_ITER_MAX_SDC = 100$, $t=[0,1e-4]$)

JFNK运行时间:

JFNK_onestep(): 35.1653 s save_results(): 0.0044 s